FUNÇÕES DIFERENCIAIS:

- Só definimos o plano tangente em $(x_0, y_0, f(x_0, y_0))$ se a função é diferenciável no ponto (x_0, y_0)

- Dizemos que f é diferenciável se $\lim_{(x,y)\to(a,b)} \dfrac{f(x,y) - \left[f(a,b) + \frac{\partial f}{\partial x}(a,b).(x-a) + \frac{\partial f}{\partial y}(a,b).(y-b)\right]}{\|(x,y)-(a,b)\|} = 0$

Onde $\|(x,y)-(a,b)\| = \sqrt{(x-a)^2+(y-b)^2}$

↳ Plano tangente

$$\boxed{z - f(x_0,y_0) = \frac{\partial f}{\partial x}(x_0,y_0).(x-x_0) + \frac{\partial f}{\partial y}(x_0,y_0).(y-y_0)}$$

Seja $f(x,y) = 3x^2y - x$. Determine a equação do plano tangente ao gráfico no ponto $(1, 2, f(1,2))$:

$f(x,y) = 3x^2y - x \qquad f(1,2) = 3.1^2.2 - 1 = 5$

* P de tg: $(1, 2, f(1,2)) = (1, 2, 5)$

$f(1,2) = 5$

* $\frac{\partial f}{\partial x}(x,y) = 6xy - 1 \Rightarrow \frac{\partial f}{\partial x}(1,2) = 6.1.2 - 1 = 11$

* $\frac{\partial f}{\partial y}(x,y) = 3x^2 \Rightarrow \frac{\partial f}{\partial y}(1,2) = 3.1^2 = 3$

$z - 5 = 11.(x-1) + 3.(y-2) = 11x - 11 + 3y - 6 \longrightarrow \boxed{11x + 3y - z - 12 = 0}$

* equação do plano tangente no ponto $(1,2)$

$z - f(1,2) = \frac{\partial f}{\partial x}(1,2).(x-1) + \frac{\partial f}{\partial y}(1,2).(y-2)$

DIFERENCIAL:

Seja uma função $z = f(x,y)$ diferenciável. A diferencial em (x,y) ponto $D(f)$, denotado por df ou dz é dado por: $\boxed{df(x,y) = \frac{\partial f}{\partial x}(x,y)\,dx + \frac{\partial f}{\partial y}(x,y)\,dy}$

$\boxed{f(x,y) \cong \frac{\partial f}{\partial x}(a,b).\Delta x + \frac{\partial f}{\partial y}.\Delta y + f(a,b)}$ * Equação para aproximação de raízes

Expressão: $(\underbrace{0{,}995}_{x})^4 + (\underbrace{2{,}001}_{y})^3 \longrightarrow$ O ponto $(0.995, 2.001)$ aplicando em $f(x,y) = x^4 + y^3$

- $\Delta x = x - a = 0{,}995 - 1 = -0{,}005$
- $\Delta y = y - b = 2{,}001 - 2 = 0{,}001$
- $f(1,2) = 1^4 + 2^3 = 1 + 8 = 9$

- $\frac{\partial f}{\partial x} = 4x^3 \longrightarrow \frac{\partial f}{\partial x}\underbrace{(1,2)}_{} = 4.1^3 = 4$; $\frac{\partial f}{\partial y} = 3y^2 \longrightarrow \frac{\partial f}{\partial y}\underbrace{(1,2)}_{} = 12$

APROXIMAÇÃO

* Agora com todos os dados:

$$f(x,y) \cong \frac{\partial f}{\partial x}(1,2).(-0{,}005) + \frac{\partial f}{\partial y}(1,2).(0{,}001) + f(1,2) \Longrightarrow (0{,}995)^4 + (2{,}001)^3 \cong -0{,}02 + 0{,}012 + 9 \cong 8{,}992$$

Exercícios:

10) Determinar a equação dos planos tangentes à superfície de f, no ponto indicado:

a) $f(x,y) = \sqrt{1-x^2-y^2}$, $P(0,0,0)$

b) $f(x,y) = \sqrt{1-x^2-y^2}$, $P\left(\frac{1}{2}, \frac{1}{2}, \frac{\sqrt{2}}{2}\right)$

c) $f(x,y) = xy$, $P(1,1,1)$

d) $f(x,y) = 2x^2 - 3y^2$, $P(1,1,-1)$

e) $f(x,y) = x\,e^{(x+y)}$, $P(1,1,f(1,1))$

11) Utilizando o conceito de diferencial, encontre um valor aproximado para:

a) $\left[(1{,}01).e^{0{,}015}\right]^7$ b) $(1{,}02)^3 + (3{,}98)^2$

$$\boxed{z - f(x_0,y_0) = \frac{\partial f}{\partial x}(x_0,y_0)\cdot(x-x_0) + \frac{\partial f}{\partial y}(x_0,y_0)\cdot(y-y_0)} \longrightarrow \text{Plano tangente.}$$

10) a) $P(0,0,1)$

$f(x,y) = \sqrt{1-x^2-y^2} \longrightarrow f(0,0) = \sqrt{1} = 1$ $\qquad u = 1-x^2-y^2 \qquad u^{\frac{1}{2}} = \frac{1}{2\sqrt{u}}\cdot u'$

$f_x(x,y) = \frac{1}{2\sqrt{1-x^2-y^2}}\cdot -2x = \frac{-x}{\sqrt{1-x^2-y^2}}$ $\qquad z-1 = \frac{-x}{\sqrt{1-x^2-y^2}}\cdot(x-0) + \left(\frac{-y}{\sqrt{1-x^2-y^2}}\right)\cdot(y-0)$

$f_y(x,y) = \frac{1}{2\sqrt{1-x^2-y^2}}\cdot -2y = \frac{-y}{\sqrt{1-x^2-y^2}}$ $\qquad$ * No ponto $(0,0,1)$ $\boxed{z=1}$

b) $f(x,y) = \sqrt{1-x^2-y^2} \qquad P\left(\frac{1}{2}, \frac{1}{2}, \frac{\sqrt{2}}{2}\right)$

$f_x(x,y) = \frac{1}{2\sqrt{1-x^2-y^2}}\cdot -2x = \frac{-x}{\sqrt{1-x^2-y^2}} = \frac{-\frac{1}{2}}{\sqrt{1-\frac{1}{4}-\frac{1}{4}}} = \frac{-\frac{1}{\cancel{2}}}{\frac{\sqrt{2}}{\cancel{2}}} = -\frac{1}{\sqrt{2}}\cdot\frac{\sqrt{2}}{\sqrt{2}} = -\frac{\sqrt{2}}{2}$

$f_y(x,y) = \frac{1}{2\sqrt{1-x^2-y^2}}\cdot -2y = \frac{-y}{\sqrt{1-x^2-y^2}} = -\frac{\sqrt{2}}{2}$

$z - \frac{\sqrt{2}}{2} = -\frac{\sqrt{2}}{2}\left(x-\frac{1}{2}\right) + \left(-\frac{\sqrt{2}}{2}\right)\cdot\left(y-\frac{1}{2}\right) \longrightarrow \boxed{z = -\frac{\sqrt{2}}{2}x - \frac{\sqrt{2}}{2}y + \sqrt{2}}$

$-\frac{\sqrt{2}}{2}\cdot x + \frac{\sqrt{2}}{4} - \frac{\sqrt{2}}{2}\cdot y + \frac{\sqrt{2}}{4}$

c) $f(x,y) = xy \qquad P(1,1,1)$

$\cdot f(1,1) = 1 \qquad \cdot f_x(x,y) = 1 \qquad \cdot f_y(x,y) = 1$

$z-1 = 1\cdot(x-1) + 1\cdot(y-1)$

$z-1 = x-1+y-1$

$z = x+y-1 \longrightarrow x+y-z-1=0$,,

d) $f(x,y) = 2x^2-3y^2, \quad P(1,1,-1)$

$\cdot f(1,1) = 2\cdot1^2-3\cdot1^2 = -1$

$\cdot f_x(x,y) = 4x \qquad (1,1) = 4$

$\cdot f_y(x,y) = -6y \qquad (1,1) = -6$

$z+1 = 4\cdot(x-1) - 6\cdot(y-1)$

$z+1 = 4x-4-6y+6$

$z+1 = 4x-6y+2$

$\boxed{4x-6y-z+1=0}$

e) $f(x,y) = xe^{(x+y)}, \; P(1,1,f(1,1))$ $\qquad \cdot f_x(x,y) = e^{(x+y)}\cdot 1 = e^{(x+y)}$

$\cdot f(1,1) = 1e^2 = e^2 \qquad u = x+y \qquad \cdot f_y(x,y) = e^{x+y}\cdot 1 = e^{(x+y)}$

$\boxed{z - e^2 = e^{(x+y)}\cdot(x-1) + e^{(x+y)}\cdot(y-1)}$

$$\boxed{f(x,y) \cong \frac{\partial f}{\partial x}(a,b)\cdot\Delta x + \frac{\partial f}{\partial y}\cdot\Delta y + f(a,b)}$$

11) a) $\left[1{,}01\cdot e^{0{,}015}\right]^7 \qquad f(x,y) = \left[xe^y\right]^7 \longrightarrow f(1{,}01,\ 0{,}015)$???

$(a,b) = (1,0)$ * Ponto próximo ao que quero.

$\Delta x = x - a = 1{,}01 - 1 = -0{,}01$

$\Delta y = y - b = 0{,}015 - 0 = 0{,}015$

$\frac{\partial f}{\partial x} = 7\cdot(xe^y)^6\cdot e^y \longrightarrow \frac{\partial f}{\partial x}(a,b)^{1,0} = 7\cdot(1\cdot e^0)^6 = 7$

$\frac{\partial f}{\partial y} = 7(xe^y)^6\cdot xe^y \longrightarrow \frac{\partial f}{\partial y}(a,b)^{1,0} = 7\cdot(1\cdot e^0)^6\cdot 1\cdot e^0 = 7$

$f(a,b) = f(1,0) = 1$

$f(1{,}01,\ 0{,}015) \cong 7\cdot -0{,}01 + 7\cdot 0{,}015 + 1 \cong 1{,}175$

b) $(1{,}02)^3 + (3{,}98)^2 \longrightarrow f(x, y) = x^3 + y^2$ $\quad P(1, 4)$ *Ponto aproximado.

$\dfrac{\partial f}{\partial x}(1,4) = 3x^2 = 3 \cdot 1^2 = 3$; $\dfrac{\partial f}{\partial y}(1,4) = 2y = 2 \cdot 4 = 8$

$f(x, y) = 3 \cdot 0{,}02 + 8 \cdot (-0{,}02) + 17 = 16{,}9$

$\Delta x = 1{,}02 - 1 = 0{,}02$

$\Delta y = 3{,}98 - 4 = -0{,}02$

$f(1,4) = 1 + 16 = 17$

*Portanto, para aproximar expressões com diferenciais, devemos seguir a formula do plano tangente e fazer aproximações com a base da expressão.

$$\boxed{f(x, y) \cong \frac{\partial f}{\partial x}(a, b) \cdot \Delta x + \frac{\partial f}{\partial y} \cdot \Delta y + f(a, b)}$$